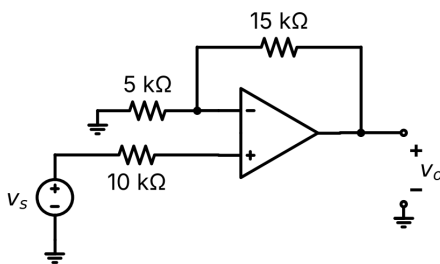


Practice sheet: The Operational Amplifier

This sheet is not marked for correctness, and AI tools are allowed. The device-free quiz is built from twins of these questions, so make sure you can do each one on your own.

- **Two-tier items** ask for an answer *and* a reason. Get both right; on the quiz each earns half the marks.
- **Predict first:** where asked, write your prediction before you calculate.
- **Confidence:** next to each answer, write S (sure) or U (unsure). This is not marked; it shows you what to revise.
- **Conventions:** every op amp is ideal: no current enters either input, and with negative feedback in the linear region the two input voltages are equal ($v_n = v_p$). The output stays between the supply voltages; outside that range the op amp saturates. Voltages are measured from the ground symbol, and resistances are in kilohms, so currents come out in milliamperes. Check every answer: KCL at the inverting input must hold, and v_o must lie between the supplies.

S1. Which configuration? (two-tier)



What is this circuit, and what is v_o/v_s ?

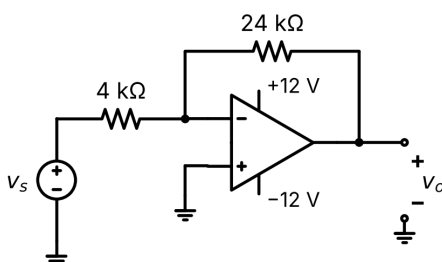
- Inverting amplifier, -3
- Noninverting amplifier, 4
- Noninverting amplifier, 3
- Difference amplifier, 3

Reason:

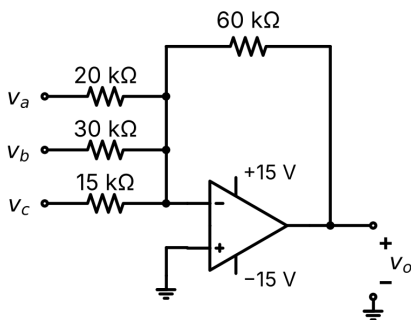
- Every op-amp amplifier inverts, because the feedback goes to the inverting input.
- The source drives the noninverting input; the feedback divider gives $v_n = v_o \times 5/(5 + 15)$, and $v_n = v_p = v_s$.
- The gain is always the feedback resistance divided by the input resistance.
- Any circuit with resistors on both inputs is a difference amplifier.

Follow-up: does the $10\text{ k}\Omega$ resistor change the gain? Why?

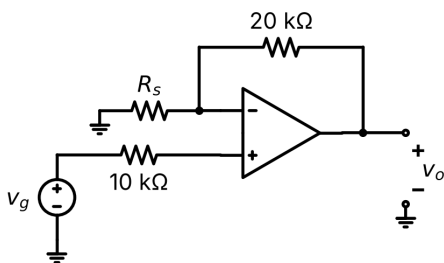
S2. The inverting amplifier. As in Nilsson Example 5.2.



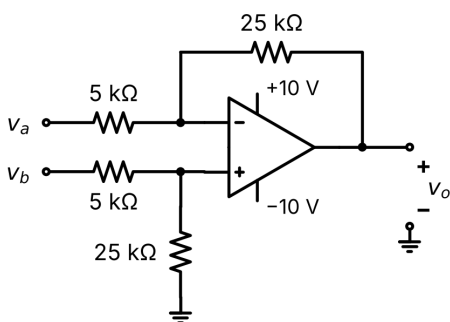
- Find v_o for $v_s = 1.5\text{ V}$.
- Find the current in the $24\text{ k}\Omega$ resistor and its direction.
- What range of v_s keeps the op amp in its linear region?
- Predict first:* what is v_o for $v_s = 3\text{ V}$?

S3. The summing amplifier. As in Nilsson Example 5.3.

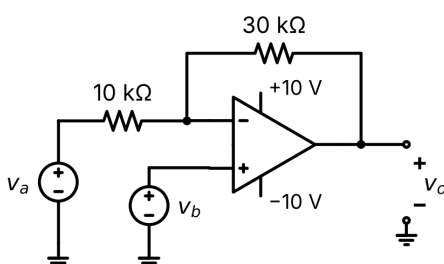
- Write v_o in terms of v_a , v_b and v_c , and find v_o for $v_a = 1$ V, $v_b = 2$ V and $v_c = -1.5$ V.
- With $v_a = 1$ V and $v_c = -1.5$ V, what range of v_b keeps the op amp linear?
- With $v_a = 2$ V, $v_b = 1$ V and $v_c = -1$ V, how large can the feedback resistor be before the op amp saturates?

S4. The noninverting amplifier. As in Nilsson Example 5.4.

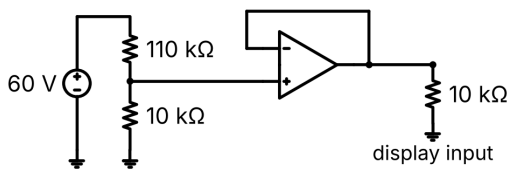
- Choose R_s for a gain of 5.
- v_g ranges from -2 V to $+2.5$ V. What are the smallest equal supply voltages ($\pm V_{CC}$) that keep the op amp linear?
- What current flows in the 10 kΩ resistor? What does that resistor do to v_o ?

S5. The difference amplifier. As in Nilsson Example 5.5.

- Name the configuration and write v_o in terms of v_a and v_b .
- Find v_o for $v_a = 2$ V and $v_b = 2.8$ V.
- With $v_b = 2.4$ V, what range of v_a keeps the op amp linear?

S6. Both inputs driven. As in Nilsson Example 5.1.

- Find v_o for $v_a = 2$ V and $v_b = 0$ V.
- Repeat for $v_a = 2$ V and $v_b = 1$ V. Why is v_n no longer 0?
- With $v_a = 1.5$ V, what range of v_b avoids saturation?

S7. A buffer for a battery monitor (predict first).

A 48 V solar battery bank can reach 60 V. A divider scales it down for a display that reads 0 to 5 V.

- Without the op amp, the display is connected straight across the 10 kΩ. *Predict first:* will it read 5 V? Then find what it reads at 60 V.
- With the op amp in place, what does the display read? Why does the display no longer load the divider?
- Where does the 0.5 mA in the display come from?

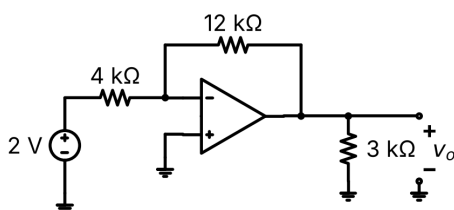
S8. Saturation (two-tier). An inverting amplifier has $R_s = 2 \text{ k}\Omega$ and $R_f = 20 \text{ k}\Omega$, with supplies $\pm 12 \text{ V}$. What is v_o for $v_s = 1.5 \text{ V}$?

- +12 V
- −15 V
- −12 V
- 0 V

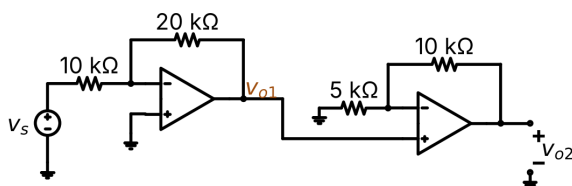
Reason:

- $v_o = -(R_f/R_s)v_s$ holds for every input.
- A saturated op amp gives zero output.
- The output cannot pass a supply voltage, so it stops at the negative supply.
- A saturated op amp flips to the opposite supply.

Follow-up: while saturated, is v_n still 0 V? Find it.

S9. The output current.

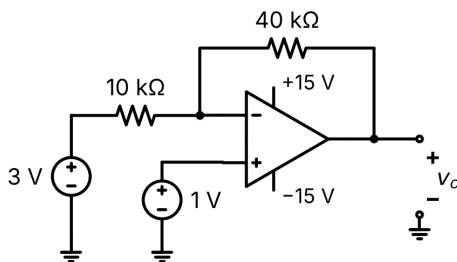
- Find v_o .
- Find the current in the 3 kΩ load and in the 12 kΩ resistor, with directions.
- Find the current at the op amp's output terminal. No current enters the inputs: why can the output carry current?

S10. Two op amps in cascade.

Both op amps use $\pm 9 \text{ V}$ supplies.

- Find v_{o1} and v_{o2} for $v_s = 0.4 \text{ V}$.
- Repeat for $v_s = 2 \text{ V}$. Which stage saturates?
- What range of v_s keeps both stages linear?

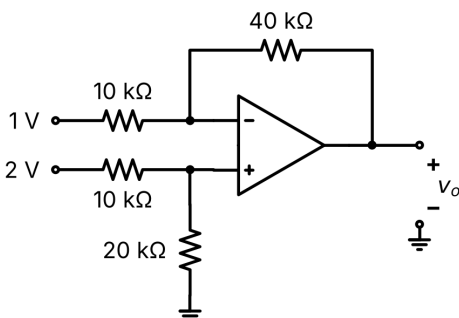
S11. Spot the error. The worked solution below finds v_o . It contains the kind of direction slip AI chat tools often make.



1. Assume the op amp is linear, so $v_n = v_p$ (this is checked in line 5).
2. $v_p = 1$ V, so $v_n = 1$ V.
3. The current from the 3 V source into the inverting node is $(3 - 1)/10 = 0.2$ mA.
4. It flows on through the 40 kΩ to the output, so $v_o = 1 + 0.2 \times 40 = 9$ V.
5. Check: 9 V lies inside ± 15 V, so the op amp is linear and the answer stands.

- a) Which is the first wrong line?
- b) Correct it and find v_o .
- c) Check your answer with KCL at the inverting input.
- d) Line 5 calls itself a check. Why does it prove nothing?

S12. Spot the error in a difference amplifier.



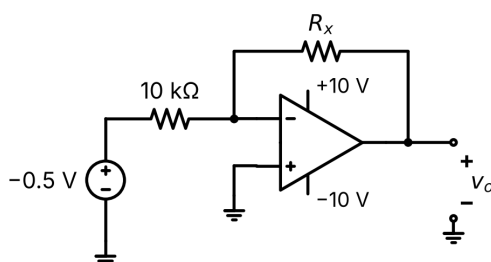
1. This is a difference amplifier, so $v_o = (40/10)(v_b - v_a)$.
2. So $v_o = 4(2 - 1) = 4$ V.
3. Check: 4 V lies inside any usual supply range.

- a) Which is the first wrong line?
- b) Find v_p , then v_o .
- c) What resistor value at the noninverting input would make line 1 correct?

S13. Design a summing amplifier. Design a summing amplifier with $v_o = -(2v_a + 5v_b)$, using a 100 kΩ feedback resistor and ± 12 V supplies.

- a) Choose the input resistors.
- b) With $v_a = 1$ V, what range of v_b keeps the op amp linear?
- c) Find the current in the feedback resistor for $v_a = 1$ V and $v_b = 1$ V.

S14. Work backwards.

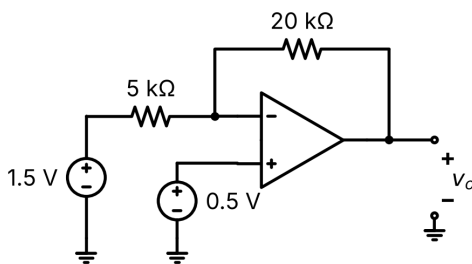


- a) Find v_o when $R_x = 150$ kΩ.
- b) How large can R_x be before the op amp saturates?
- c) What happens to v_o if R_x is made larger still?

S15. From a sensor to a microcontroller. As in Nilsson's Practical Perspective for this chapter. A temperature sensor gives 10 mV per °C, so 0 to 100 °C gives 0 to 1 V. The microcontroller's analog input reads 0 to 5 V.

- What gain maps 0–1 V onto 0–5 V? Design a noninverting amplifier for it with a 40 k Ω feedback resistor.
- The amplifier's supplies are 0 V and +5 V. What does it output at 120 °C, and what temperature will the microcontroller report?
- A second sensor reads -20 °C to 80 °C, that is, -0.2 V to 0.8 V. Find K and L in $v_o = Kv_s + L$ that map this range onto 0–5 V.

S16. Check an answer without solving again.



Two classmates report v_o . Student 1: $v_o = -3.5$ V. Student 2: $v_o = 4.5$ V. Use $v_n = v_p$ and KCL at the inverting input to decide which student is right.